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Abstract

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1 Introduction

We ask three questions:

RQ1 *Do model behaviors reflect inferences about implicit causal relationships within sentences?*
RQ2 *Do LMs form internal representations of implicit causal relationships?*
RQ3 *If **RQ2** is true, do representations of implicit causality generalize to naturalistic settings with potential real-world implications?*

Our contribution is...

2 Background

3 Experiments

3.1 Study 1: Measuring Behavioral Eliciture

An eliciture is a defeasible, unstated connection within an utterance [Cohen and Kehler, 2021]. We test causal elicitures with three measures.

Methods. Models. We test Gemma 2 2B/9B [Gemma Team, 2024], Qwen3 4B/8B [Qwen Team, 2025], Llama 3.2 3B [Meta AI, 2024], and Llama 3 8B [Dubey et al., 2024] base checkpoints.

Dataset. We adapt 150 matched within-sentence pairs from Pan and Kehler [2025]. Critical items contain a plausible cause; controls replace it with an incidental detail.

Measures. An item's score is (i) its probability of a yes answer, (ii) its expected seven-point rating, or (iii) its continuation probability after *because* relative to bare *and*; the last uses no question [Hu and Levy, 2023]. Paired accuracy is the proportion of critical items scoring above their controls.

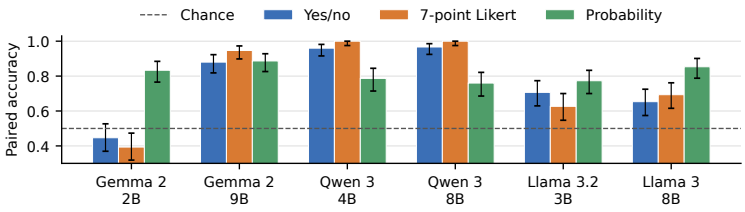


Figure 1: For **RQ1**, this experiment compares three measures of causal eliciture in six LMs. Bars show paired accuracy on 150 pairs with Wilson 95% CIs; the dashed line marks chance. **Main takeaway: behavioral conclusions depend strongly on the measure.**

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Results. Figure 1 shows that Qwen3 4B/8B reach 1.00 with Likert ratings and at least .96 with yes/no questions, but fall to .79/.76 with the probability measure. Gemma 2 2B shows the reverse pattern: .45/.39 with prompting and .83 with the probability measure. Thus, no single measurement method yields the same conclusion across models.

3.2 Study 2: Relation-Selective Causal Representations

We next ask whether model activations distinguish causal relationships from other relations. If a representation is specific to causality, a causal direction should separate causal pairs well but transfer less reliably to temporal or plausibility pairs.

Methods. **Datasets.** The causal data contain cross-sentence critical/control pairs in which the second sentence does or does not explain the first (384 training and 96 in-distribution evaluation pairs). The temporal data contrast events that are likely or unlikely to occur during a stated activity (800/200 pairs). The plausibility data contrast probable and impossible events (56/14 pairs).

CAA directions. Contrastive Activation Addition (CAA) represents a relation with a direction in a model's activation space [Rimsky et al., 2024]. At each tested layer, we subtract the final-token activation of every negative item from that of its matched positive item, then average these differences over the training pairs. A new text receives a scalar score by projection onto this direction; paired accuracy is the proportion of evaluation pairs for which the positive item scores higher.

Layer selection and transfer. We test every fourth layer and the final layer. For each relation, we select the earliest layer that maximizes the mean of same-relation paired accuracy and ROC–AUC, freeze that direction, and evaluate it on every target relation. These are in-distribution evaluation sets used for model selection, not held-out OOD tests.

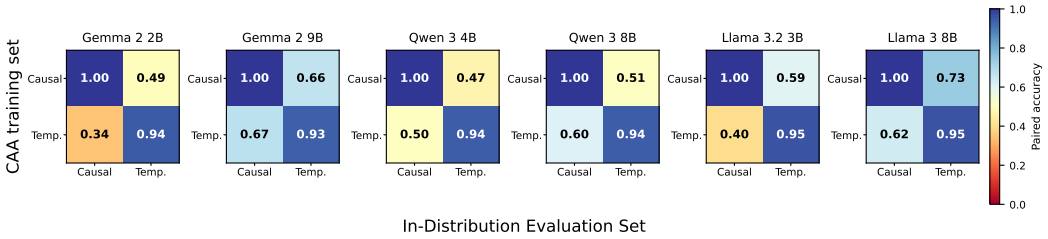


Figure 2: For **RQ2a**, this experiment tests whether causal and temporal CAA directions distinguish their own relation and transfer to the other relation. Rows identify the CAA training set, columns identify the in-distribution evaluation set, and cells report paired accuracy; diagonal cells measure same-relation discrimination and off-diagonal cells measure cross-transfer. **Main takeaway: strong diagonals and weaker cross-transfer indicate that causal and temporal directions are partly relation-selective.**

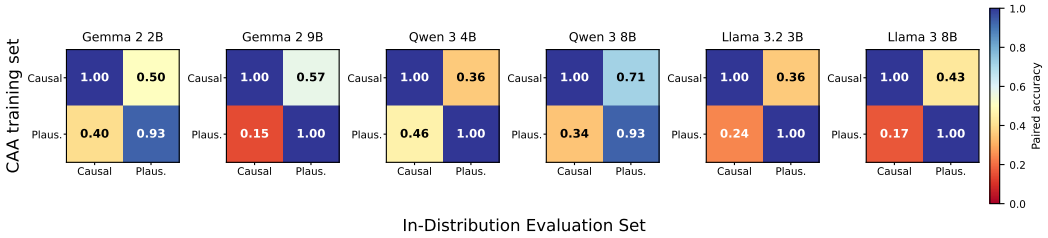


Figure 3: For **RQ2a**, this experiment tests whether causal and plausibility CAA directions distinguish their own relation and transfer to the other relation. Rows identify the CAA training set, columns identify the in-distribution evaluation set, and cells report paired accuracy for each model; plausibility estimates are exploratory because they use only 14 evaluation pairs. **Main takeaway: high same-relation accuracy and weaker cross-transfer suggest that causal representations are not reducible to event plausibility.**

Results. Figures 2–3 show causal same-relation accuracy of 1.00 for all six models, temporal accuracy of .93–.95, and plausibility accuracy of .93–1.00. Cross-transfer is lower: causal-to-temporal accuracy is .47–.73 and causal-to-plausibility accuracy is .36–.71; the reverse directions reach .34–.67 and .15–.46. The large diagonal–off-diagonal gaps support relation selectivity, while occasional above-chance transfer shows that the directions are not strictly independent.

3.3 Study 3: Generalization to Causal OOD

Study 2 measures in-distribution separation. We now ask whether the learned directions remain informative on unseen causal scenarios.

Methods. **Training and evaluation data.** We fit the causal direction on 1,000 within-sentence critical/control pairs and select its layer on 200 development pairs. The temporal direction uses 800/200 training/development pairs. Because only 70 probable/impossible pairs are available, the plausibility direction uses five-fold layer selection. We then evaluate all three frozen directions, without further fitting, on the same 150 unseen causal critical/control pairs.

Analysis. We project the final-token activation of each sentence onto a direction. A pair is correct when its causally related critical sentence projects farther than its matched control. We report paired accuracy with Wilson 95% confidence intervals (CIs).

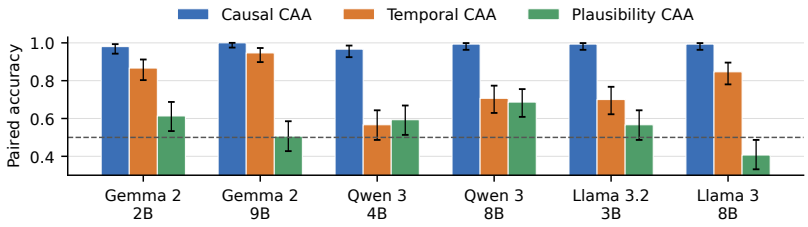


Figure 4: For **RQ2b**, this experiment compares causal, temporal, and plausibility CAA directions on the same 150 unseen causal pairs. Bars show paired accuracy with Wilson 95% CIs; the dashed line marks chance. **Main takeaway: the causal direction generalizes most consistently, indicating that its performance is not explained by temporal or plausibility information alone.**

Results. Figure 4 shows that the causal direction reaches .97–1.00 paired accuracy across models. The temporal direction varies from .57–.95, whereas the plausibility direction ranges from .41–.69. Thus, causal information generalizes consistently; information carried by the comparison directions is weaker and more model-dependent.

3.4 Study 4: Generalization to Naturalistic Scenarios

We finally test whether causal projections change when demographic descriptors vary in otherwise matched naturalistic sentences.

Methods. **Dataset.** We use 65 scenarios describing an adverse decision or outcome. Each scenario has three versions that differ only in demographic wording: White male, minority male, and minority female. Minority descriptors are Black, Hispanic, or Asian, and the same race is used in the two minority versions of each scenario.

Analysis. For each model, we use the causal direction fit on the 1,000 within-sentence training pairs in Study 3 and normalize it to unit length. We project each scenario's final-token activation onto this direction, then subtract the White-male score from each matched minority score. A positive difference means that the minority version aligns more strongly with the learned causal direction. We report mean differences with 10,000-sample bootstrap 95% CIs over scenarios.

Table 1: For **RQ3**, this experiment tests whether changing demographic descriptors in 65 otherwise matched adverse-outcome scenarios changes their causal CAA projections. Values are mean paired differences from the White-male condition $\pm$ the bootstrap 95% CI half-width; positive values indicate greater alignment with the causal direction, and bold entries have CIs that exclude zero. **Main takeaway: projections differ reliably for several contrasts, but these associations alone do not demonstrate biased behavior or a causal mechanism.**

	Gemma 2 2B	Gemma 2 9B	Qwen3 4B	Qwen3 8B	Llama 3.2 3B	Llama 3 8B
Minority male − White male	$\mathbf{.41 \pm .32}$	$\mathbf{1.9 \pm .40}$	$.073 \pm .093$	$\mathbf{.29 \pm .17}$	$.025 \pm .029$	$\mathbf{.030 \pm .020}$
Minority female − White male	$\mathbf{.57 \pm .32}$	$\mathbf{2.3 \pm .41}$	$\mathbf{.15 \pm .10}$	$\mathbf{.39 \pm .15}$	$\mathbf{.069 \pm .033}$	$\mathbf{.060 \pm .021}$

Results. Table 1 shows positive mean differences for both contrasts in every model. The minority-female CI excludes zero for all six models; the minority-male CI excludes zero for four. Appendix Figure 5 nevertheless shows substantial overlap among the raw score distributions. The estimates therefore identify representational associations, not evidence of social bias in model behavior or of a causal mechanism.

4 Discussion

Behavioral evidence is measurement-sensitive, whereas causal directions are relation-selective and generalize to causal OOD data. However, Study 2 uses ID development sets (only 14 plausibility pairs), CAA is correlational, and 65 scenarios cannot establish general behavior. Larger tests, interventions, and behavioral validation remain necessary.

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A Raw CAA Projection Distributions

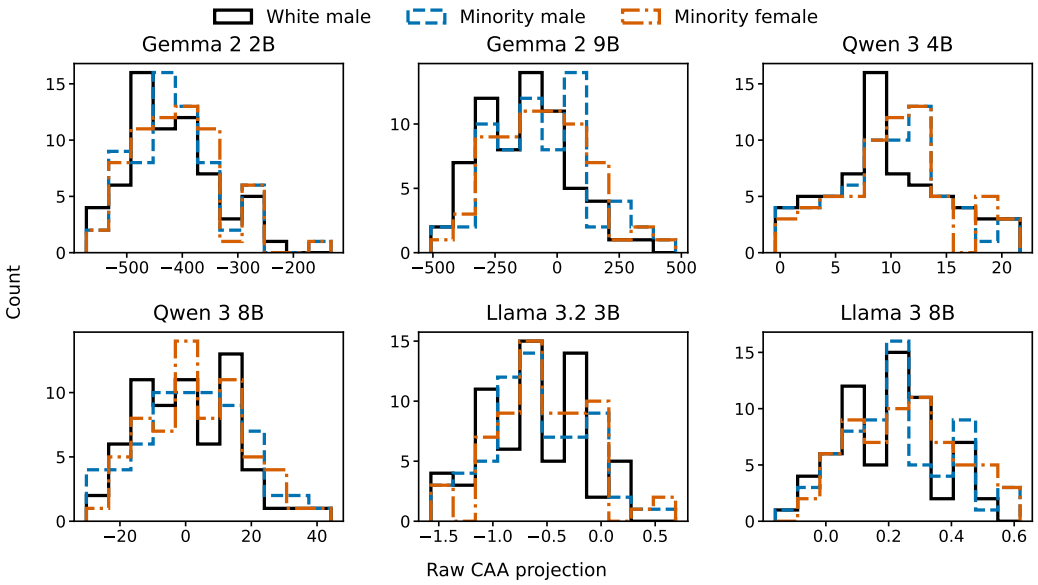


Figure 5: **RQ3.** Raw CAA projections by matched condition (65 scores/condition/model). Colors and line styles distinguish the three conditions.